Monte Carlo Simulations and Financial Options

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Introduction

Monte Carlo integration is a statistical computing method used to approximate integrals based on random sampling. Monte Carlo simulations use sampling from an appropriate distribution, and a calculation of the sample mean of a given quantity to approximate the corresponding population mean which is supposed to be the correct answer. If we assume a problem is given by an integral I, we can perhaps write this as

$$I = \int f(y)g(y)dx \tag{1}$$

where $f(x)$ is a distribution and the integration region is the corresponding support. We then are tasked with estimating $\mathbb{E}(g(y))$ which can be done with the sample mean.

The simulations achieve many results from the samples drawn and then averages those results to obtain a final estimate. These multidimensional simulations can then be used to model various outcomes that might be difficult otherwise with numerical integration. These simulations can be used in many different areas and fields including finance, engineering, supply chain and many others.

In order to apply what we have learned regarding Monte Carlo methods and simulations, our group was able to identify an article within statistical literature that applies Monte Carlo to financial options. "Options: A Monte Carlo Approach" was published in 1977 by Phelim P. Boyle in the Journal of Financial Economics. While the article is 44 years old, much of the paper can still be applied to today's financial markets and statistical approaches.

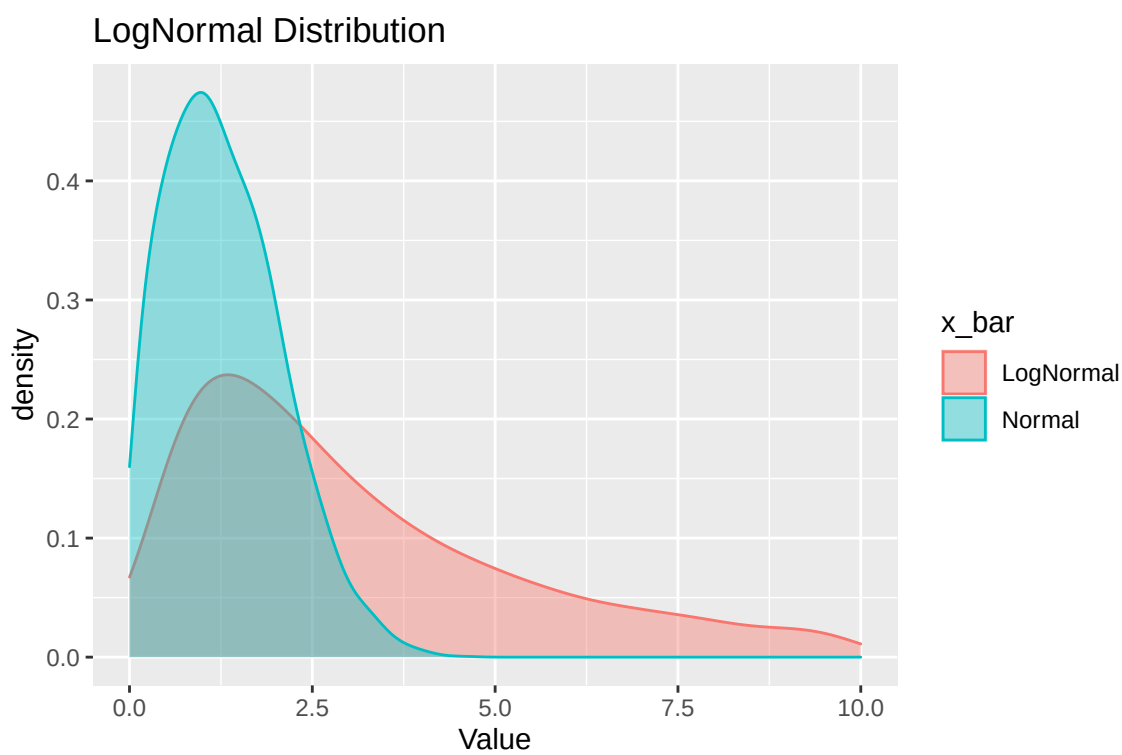
Financial options are a very powerful and influential tool used heavily in financial markets. An option is a contract that is valued based on a different underlying asset or security like a common stock. A single options contract represents 100 shares of the underlying asset it is representing. Because they represent 100 shares of the underlying asset, options are more risky than common stock as the price of the option moves faster. The piece of the paper we wish to replicate simulates the process of generating returns on the underlying assets and then deriving the value of the option.

Financial Terminology

In order to understand the wording and how our simulation works, first some financial terms need to be defined.

- Common Stock: security ownership in a particular company
  - ex: common stock of Apple is ~ $165
- Dividend: A sum of money paid regularly by a company to its shareholders, its important to note that not all stock give dividends to the companies investors. Dividend's are usually used as motivation to get more people to invest in a company.

- Inflation Rate: rate at which the value of a currency is falling and the general level of which prices for goods and services is rising. Inflation rate is a sign of economic growth and necessary to prevent stagflation. If interest-rates are too high this leads to hyper-inflation. The ideal intrastate for an optimal growing economy is around 2%
- Strike Price: The set agreed price at which the underling stock in the contract can be bought or sold when it is exercised.
- Option's Premium: the current market price of an option contract which is based on the stocks current value, strike price, inflation rate and risk free rate. The premium is what you pay someone to sell you an option contract. You are essentially paying someone else an upfront payment for them to take the risk of holding 100 shares of a stock for each contract.
- Treasury Yield: yield of a US Treasury Bond matching investment duration
- Risk Free Rate: represents the interest an investor would expect from a risk-free investment over a specific time
  - RFr = Inflation Rate - Treasury Yield
- LogNormal Distribution: distribution of logarithmic values from a related normal distribution
  - Allow for only positive variables whereas the normal distribution can allow for negative variables
  - More useful in finance because the log normal curve can better identify the compounding return a stock can achieve over a long period of time
  - As shown in the plot below, the longer right tail of the LogNormal distribution can increase stock returns due to a larger compounding effect compared to a normal distribution.



Overview of Options

Options are a very powerful and influential tool used heavily in financial markets. Investors and investment institutions, like hedge funds and asset managers, use options for income, to hedge risk and to speculate

about future market movements. Options are more risky than common stock as the price moves faster, and in many cases with higher volatility.

So, what is an options contract? It is a contract that its value is based on an underlying asset or security like a common stock or bond. It is the most common form of a derivative, meaning its price is linked to the price of another underlying asset. A single options contract represents 100 shares of the underlying asset it is representing. The contract gives the buyer the right to buy the underlying asset at a specific price on or by a certain date. There are two types of financial options: European and American. European options can only be exercised, or right to buy the underlying asset, at the time of expiration and American options can be exercised at any time before the options expiration. Again, an options contract gives the buyer the right to buy the underlying asset at a specific price on or by a certain date and exercising the contract just means you finish the transaction and buy the underlying asset at the exercise/strike price.

Valuing options is essentially about determining the probabilities of future prices for the underlying asset. The more likely a stock price is to reach the strike price, the more expensive the options premium will be. As the time to maturity, or time until the option expires, decreases, the less valuable the option will become. Prior to the methods listed in this paper and the Black-Scholes Model which this paper used to derive their method, there was no standard method for determining option prices.

History

The purpose of Boyle's paper was to show that Monte Carlo simulations provide another option of calculating numerical solution to option valuation and option valuation problems. A few other methods were introduced around the same time as Boyle's paper. A paper by Black and Scholes in 1973 detailed how option values would be obtained when market equilibrium conditions exist and Cox and Ross in 1976 showed that the value of a European call option can be obtained by discounting the expected maturity value of the option at the risk free rate. Boyle's Monte Carlo method is simpler and more flexible compared to the other methods as it can be changed and modified to use different processes that govern the underlying stock returns and can change depending on the stock. This method is a good supplement to the other two methods detailed above and each method has it's own advantages and disadvantages.

Methodology

Boyle's work took off where Cox and Ross (1977) stopped, deriving the expectation of the ratio of stock prices from the Black Scholes model,

$$\mathbb{E}(\frac{S_T}{S_t}) = e^{r(T-t)} \tag{2}$$

In our case, we're looking at discrete quarters, so

$$\mathbb{E}(\frac{S_{t+1}}{S_t}) = e^{r} \tag{3}$$

The black scholes model tells us this ratio is distributed lognormally, so using the fact that the expectation of a log normally distrubuted random variable is $e^{\mu^2+\sigma^2}$ where μ and σ correspond to the underlying normal random variable generating the lognormal distribution, we get

$$S_{t+1} = S_T e^{r-\sigma^2+\sigma\tilde{x}} \tag{4}$$

from

$$\mathbb{E}(\frac{S_{t+1}}{S_t}) = e^{(r-\sigma^2)+\sigma^2}. \tag{5}$$

| In each cycle, stocks are assumed to take on a background random interest given by normal distribution with 0 mean and variance σ defined as the volatility of the stock. The goal of our simulation is to calculate the value of the option. This is simply how much the option could pay out if exercised. We take the $\max(S_T - E, 0)$ as this value where (4) is iterated over each quarter till the period to maturity is reached. In the actual calculation, the method accounted for the dividend paid out each quater as well, i.e. $S_{t+1} - D$ after each iteration. If it were the case, for a particular trial, that this value creeped below zero, the method outlined starting at the exact quarter where that occured and finishing out the trial getting the period to maturity. The monte carlo calculation comes from the fact that calculating the final option price for a trial accounts to the function which we mean to take the expected value of in pricing the option. Therefore, the sample mean of the trials for a particular set of conditions represents a quantifiably accurate pricing for the option.

Results

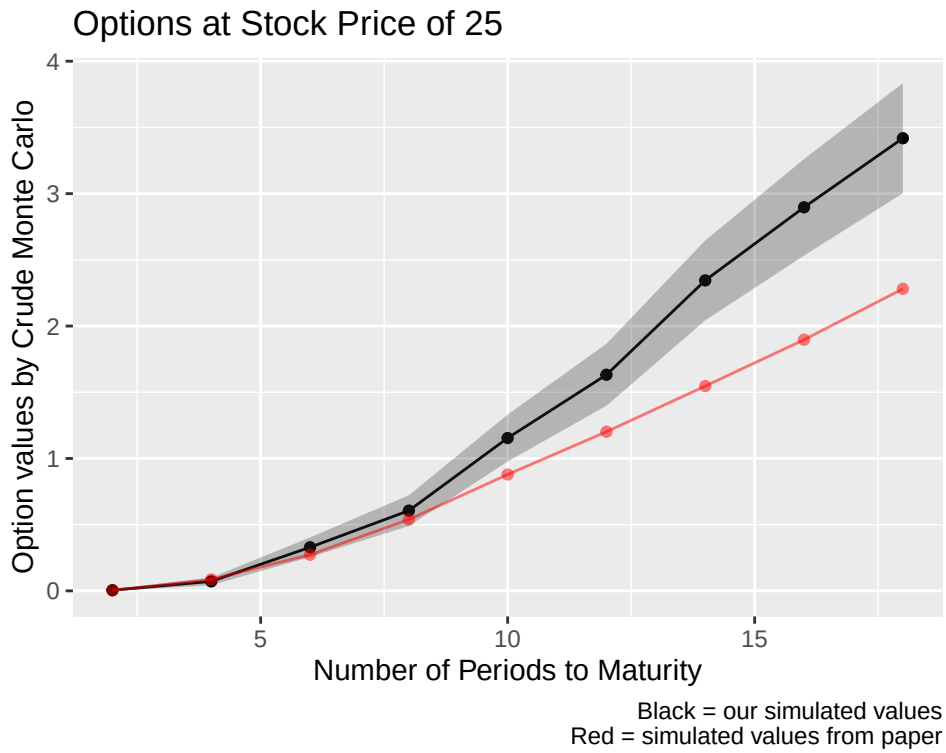
Shown below are the results for our simulation with the following values:

- Trials = 5000
- Periods to Maturity = 2, 4, 6, 8, 10, 12, 14, 16, 18, 20
- stock_prices = 25, 50, 75
- Exercise Price = 50
- Risk Free Rate = 0.015
- $\sigma^2 = 0.025$
- Dividend = 0.25

These values are the same as the simulations described in the paper.

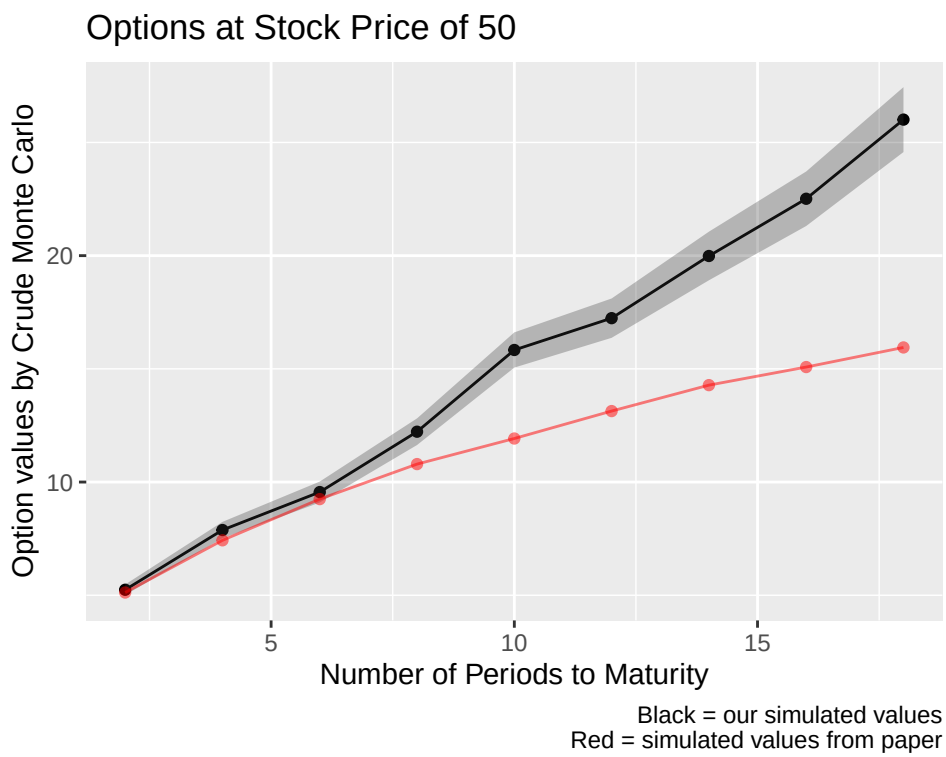
Stock Price = $25

Periods_til_maturity	Option_values	Standard_deviation	lower_95_CI	upper_95_CI
2	0.0052	0.0032	-0.0012	0.0116
4	0.0721	0.0145	0.0432	0.1010
6	0.3293	0.0372	0.2550	0.4037
8	0.6061	0.0581	0.4899	0.7223
10	1.1543	0.0887	0.9769	1.3317
12	1.6322	0.1170	1.3982	1.8662
14	2.3450	0.1510	2.0430	2.6470
16	2.8969	0.1823	2.5324	3.2614
18	3.4185	0.2080	3.0025	3.8346



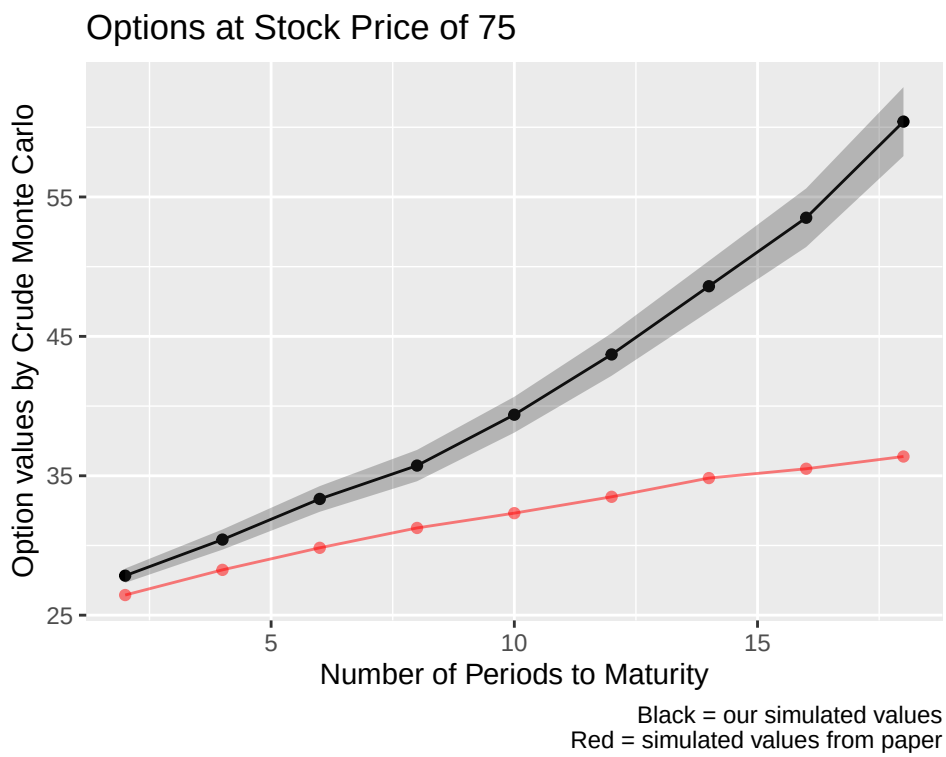
Stock Price = $50

Periods_til_maturity	Option_values	Standard_deviation	lower_95_CI	upper_95_CI
2	5.2420	0.1162	5.0096	5.4745
4	7.8808	0.1807	7.5194	8.2421
6	9.5576	0.2318	9.0941	10.0211
8	12.2239	0.2946	11.6347	12.8131
10	15.8330	0.3890	15.0551	16.6109
12	17.2389	0.4345	16.3699	18.1079
14	19.9847	0.5370	18.9108	21.0587
16	22.5127	0.6021	21.3084	23.7169
18	26.0087	0.7172	24.5742	27.4432



Stock Price = $75

Periods_til_maturity	Option_values	Standard_deviation	lower_95_CI	upper_95_CI
2	27.8347	0.2566	27.3215	28.3479
4	30.4227	0.3623	29.6981	31.1472
6	33.3373	0.4612	32.4149	34.2596
8	35.7313	0.5618	34.6077	36.8548
10	39.3815	0.6438	38.0939	40.6691
12	43.7017	0.7641	42.1735	45.2298
14	48.5979	0.9073	46.7833	50.4125
16	53.5099	1.0513	51.4074	55.6125
18	60.4049	1.2368	57.9313	62.8785



In each plot, there are clearly some differences between our simulations and the simulation results listed in the paper. Around 1977, the time in which Boyle published his paper, the inflation rate around 7% and the treasury yield was around 8% and climbing. These numbers are important because the final option price is discounted by the rate of the risk free rate, inflation rate - treasury. Today, the inflation rate is around 6% and the treasury yield is around 2%. The differences is these numbers, over a long period of time, can lead to bigger differences in our simulated values and the Boyle's simulated values.

In all, Boyle's method has withstood the test of time and clearly can still be applied to options with current numbers. The flexibility and simplicity of the Monte Carlo method makes Boyle's method a great way of simulating option prices and values. With financial options being such a large part of the overall financial market, hedge funds and investors will continue to use methods like Boyle's Monte Carlo simulations to determine the best potential investments.

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